ALIoT Smart Lab - Progress Report

17 October 2026 · Portal: **http://192.168.1.200** · branch `offline-router-door-photos` (not committed, not pushed)

Summary. Today's portal work is deployed and checked on the running portal:

- the lab images for every lab;
- a speed pass that found and fixed four real causes of lag;
- project questions to the AI agent answered in under half a second instead of about 100 s;
- UR5e master controls (exact Home pose, global STOP, recovery), kept in DRY RUN so nothing is sent to the robot;
- approved actions on the lab PCs.

Everything that needs the robots, camera 1 or the lab PCs physically is ready for testing at the lab tomorrow. Status labels: **READY** = tested here; **READY FOR PHYSICAL VALIDATION** = built and tested offline, the lab test is still needed; **BLOCKED** = waiting on something outside the code.

1. Status by component

Component	Done	Tests / actual result	Status	Remaining / next action
Lab images (all labs)	Card, page header and gallery images for PLC, Production, Materials, Automotive, PCB, Microcontrollers, Power, Design and Fashion. The gallery lists only the equipment you confirmed (e.g. Materials: tensile, bending, hardness; Power: no SCADA item). Images are labelled as illustrations.	All images load from the live portal (HTTP 200); 10/10 card images on /labs rendered with no broken images.	**READY**	Swap in real photos of your own equipment when you have them.
Portal speed	Four root causes fixed (section 2): internet check moved off the request path, face-server and ESP32-CAM checks moved to the background, larger database pool, identical requests shared.	Measured on the live portal (section 2). Backend suite: 466 passed.	**READY**	Repeat `e2e/perf_routes.mjs` on the lab network.
AI agent	Lab-data questions answered straight from the database; project and technical questions answered from the project's own documents and code, with sources and no model. Only the changed parts of the knowledge index are rebuilt.	Project questions 100-110 s → 0.12-0.45 s; lab-data questions 30-100 ms (section 3).	**READY**	A full written answer still uses the CPU model (~1-2 min) in the Project knowledge tab.

UR5e master controls	Home to the verified pendant pose; Home is confirmed only when the measured joints are within 0.5° (60 s timeout); global STOP for any staff member, which latches and blocks Home; admin recovery that restarts nothing; clear "not an emergency stop" notice; audit log entries; "driven from" display (pendant / remote / hand-guiding).	29 robot control tests, 37 robot monitor tests. Live page shows the panel, the STOP button and BLOCKED (no joint feedback, because the robot is not connected).	**READY FOR PHYSICAL VALIDATION**	Commissioning checklist (section 5). `ROBOT_CONTROL` stays `dry_run` until you sign off.
FANUC ER-4iA	Its own adapter: read-only position monitor and twin. No UR commands or poses reused.	Control reports UNAVAILABLE with the reason.	**BLOCKED**	Confirm which licensed remote interface the controller has and validate the HOME program on the pendant.
Camera 1 (SECCAM_LAB01, MC210)	Service running on port 5100.	Service healthy; no picture because the camera is at the university.	**READY FOR PHYSICAL VALIDATION**	Check its stream on the lab network (192.168.1.155).
Camera 2 (SECCAM_LAB02, MC200)	Service running on port 5101.	Live: picture streaming, 7 fps, analysis running, no errors.	**READY**	Mount it diagonally at the lab, then run the floor calibration and person-crossing tests.
Lab PCs	Health monitoring (CPU, RAM, disk, network, signed-in user), lock/unlock, view-only screen preview with an on-screen notice, messages, plus new actions: open/close an approved app, lock, log off, restart, shut down. Each action shows queued → executing → completed / failed / expired, from the agent's own report; all audited.	13 equipment tests; agent refuses unknown apps and actions.	**READY FOR PHYSICAL VALIDATION**	Install `tools/pc_agent` on each lab PC (needs IT approval). Wake-on-LAN is not available (needs BIOS/network-card support).
Services	Docker had stopped (laptop restart). Restarted the stack and ran `launch.bat`.	Database, backend, frontend, n8n, face server, both cameras, knowledge service and Ollama all running; UR5e monitor running.	**READY**	-

2. Portal speed - causes found and before / after

Cause found (measured)	Fix
`/api/system/status` checked internet access inside the request: two 1 s timeouts on the offline router, while holding a database connection (2.1 s).	Check runs in a background thread; the request answers from the last result.

Face-server and ESP32-CAM checks ran inside the request every 15 s with a 1 s timeout: 1.0 s spikes on `/api/devices` and `/api/labs/{id}` while the camera is off.	Checks refresh in the background and serve the last known result.
Database pool of 5 (+10) connections for about 20 parallel requests from the start page.	Pool of 20 (+20).
Identical requests fired together (`/summary` ×2, `/system/status` ×3), queuing behind the browser's 6-connection limit.	Identical GET requests in flight (or within 1 s) are shared; write requests are never shared.

Measurement (live portal, laptop over LAN, Chrome)	Before	After
Sign-in → app ready	3.6 s	0.95 s
`/api/system/status` on the start page	2.1 s	0.17 s
`/api/devices` , 40 calls in a row	1.0 s spikes	18-45 ms, no spikes
Full page reload, median: /admin/devices · /admin/equipment · /labs/1	2.1 · 1.9 · 1.3 s	0.91 · 0.91 · 1.05 s
Sidebar navigation (median): page shown · data loaded	not measured	80-130 ms · 140-200 ms

The full-reload figures include the measuring tool's 0.5 s wait for the network to go quiet. `/admin/security` streams video continuously, so it never goes quiet and isn't comparable. Scripts: `frontend/e2e/perf_routes.mjs` and `perf_nav.mjs` ; raw results in `docs/reports/perf_before.json` and `perf_after.json` .

3. AI agent - response times

Question (examples)	Path	Before	After
"which labs are free now", "what is offline", "open issues", "who entered LAB_01 today"	Answered directly from the database	30-100 ms	30-100 ms
"how does the door controller decide to open the door"	Project documents and code (cited)	109 s	0.15 s
"what is the UR5e home position"	Project documents and code (cited)	102 s, wrong ("nothing recorded")	0.45 s, cites ROBOT_CONTROL.md
"which pin is the reed switch on", "how does the camera tracking work"	Project documents and code (cited)	~100 s	0.13 s

The fast answers quote the matching passages with their file names and don't interpret them. A written, model-generated answer is still available in the Project knowledge tab; it runs on the CPU model, so it takes longer.

4. Tests run today (all passing)

- Backend: **466 passed** (full suite), then the robot-control and equipment tests again after the last changes (29 + 13 passed).
- Tools: robot monitor **37**, knowledge **40**, security vision **163** passed.
- Frontend: TypeScript check clean; production build OK; live browser check of /robots/10, /labs and /labs/8. The only console messages were the browser's standard COOP-header notice on plain HTTP.

5. Checklist for tomorrow at the lab (in order; stop at the first failure)

1. **Network:** laptop on Smart_Lab (192.168.1.200). The portal opens on the lab PCs and phones. Run `launch.bat` and check that every window starts.
2. **UR5e link:** cable connected, robot at 192.168.1.102, laptop pings it. The portal's Robots page shows UR5e ONLINE with joint values. *Expected: twin moves when you jog on the pendant; "Driven from: Teach pendant".*
3. **UR5e Home check:** run the pendant's own Home. *Expected: portal panel shows **Home reached** (every joint within 0.5°).*
4. **Clearance:** with the operator, check the joint-space path to Home from typical poses (ceiling mount).
5. **Remote mode** (operator decides): pendant Settings → System → Remote Control. In Local mode a portal Home must be **refused**.
6. **Sign-off → live:** set `ROBOT_CONTROL=live` in `.env` and restart only the backend. Operator's hand on the E-stop: STOP → latched → admin recovery; then Home from a nearby pose. *Expected: HOMING → HOME_REACHED from feedback.* Set it back to `dry_run` after the session unless you decide otherwise.
7. **FANUC:** confirm the controller's licensed remote option (e.g. socket messaging or PNS/UOP signals) and write HOME on the pendant. Until then it stays monitoring only.
8. **Camera 1 (MC210):** check the stream in Security. **Camera 2:** mount it diagonally, run `floorcal.py`, then walk-through tests (one person, two people crossing). Note the measured delay.
9. **Lab PCs** (with IT approval): install `tools/pc_agent` on one PC. Check it appears in Lab Equipment → Robots & PCs; test View screen (notice appears on the PC), Message, Open app → Notepad, Lock now. Test restart only when nobody is working on that PC.
10. **Wi-Fi:** the router channel is now on Auto. Re-measure the ESP32-CAM ping and face-enrolment time on site.

6. Still outstanding

Item	Why it is open
UR5e live Home / STOP	Needs the robot on the lab network, Remote mode, a clearance check and your sign-off (CLAUDE.md rule: nothing that moves a robot is sent before that).
FANUC control	Controller remote interface and licence not yet identified; HOME program not yet validated.
Safety-rated remote E-stop	Not possible in software; it would need safety I/O hardware wired to the cell. The portal STOP is a software stop only.
Lab PC agents, Wake-on-LAN	Needs installation on the lab PCs (IT approval); Wake-on-LAN needs BIOS/network-card support.
Camera 1 picture, two-camera floor calibration	Camera is at the university; camera 2 still needs mounting there.

Commit	All changes are uncommitted on `offline-router-door-photos`; files are backed up in `backups/`. Say "commit" when you want a commit (nothing is pushed).
Not started this round	Reports centre (CSV/PDF per subsystem), cross-system incident timeline, inventory discrepancy view.

7. Files changed today (main ones)

- **Backend:** `app/api/routes/system.py`, `app/services/devices.py`, `app/db/session.py`, `app/api/routes/ai.py`, `app/services/robot_control.py`, `app/services/ur_script.py` (new), `app/api/routes/robots.py`, `app/api/routes/equipment.py`; tests `test_monitor.py`, `test_robot_control.py`, `test_robots.py`, `test_equipment.py`.
- **Frontend:** `lib/api.ts`, `components/robot/RobotControlPanel.tsx`, `components/robot/RobotTwinPanel.tsx`, `components/PcActions.tsx` (new), `components/NetworkedEquipment.tsx`, `components/LabCard.tsx`, `pages/LabDetail.tsx`, `components/EquipmentGallery.tsx`, images under `public/assets/labs/`; scripts `e2e/perf_routes.mjs`, `e2e/perf_nav.mjs`.
- **Tools and docs:** `tools/pc_agent/pc_agent.py` and its README, `tools/robot_monitor/control_source.py`, `docs/ROBOT_CONTROL.md`, `docs/PROGRESS_CHECKPOINT.md`.